MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, MODULAR, TYPE IV,
CONNECTOR, 100 PIN, OFFSET CONTACT TAILS ON .050 CENTERS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the connector described herein shall consist of this specification and the latest issue of MIL-C-28754.

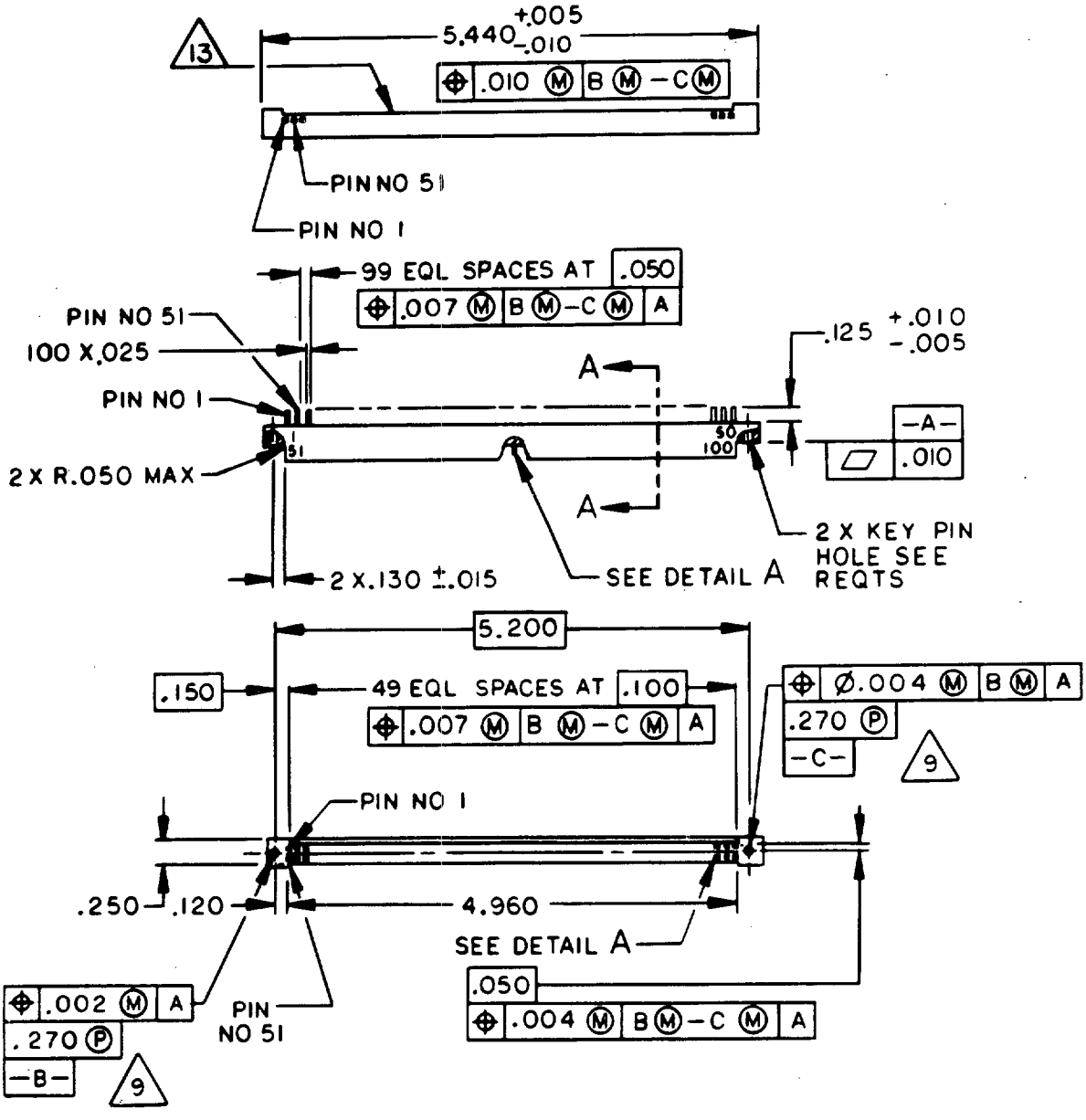
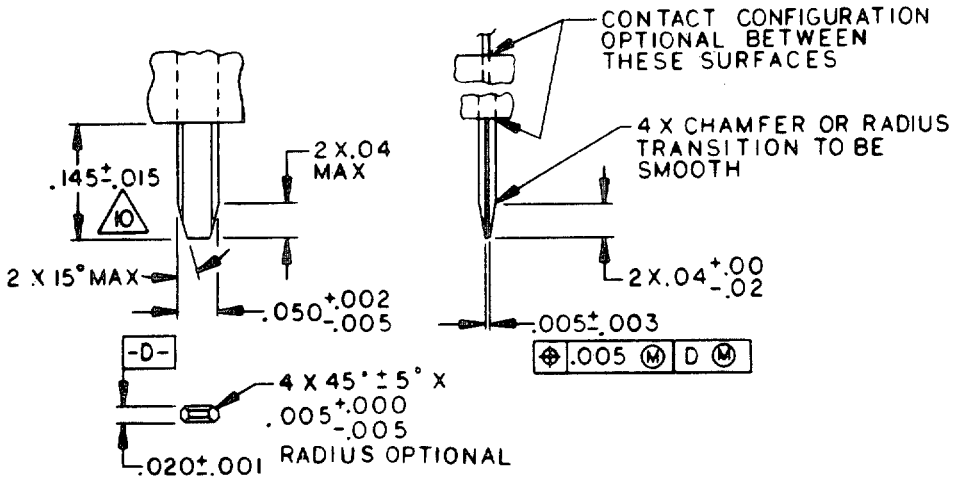


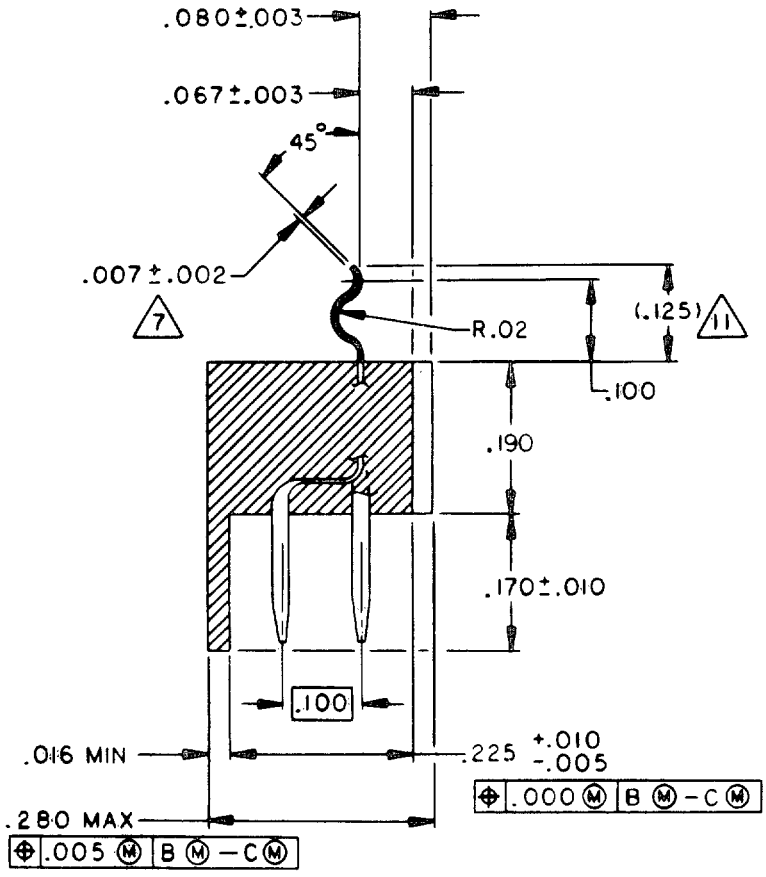
FIGURE 1. Dimensions and configurations

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DETAIL A
100 X



SECTION A-A

FIGURE 1. Dimensions and configurations - continued

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INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.001	.03	.010	.25	.050	1.27	.130	3.30	.250	6.35
.002	.05	.015	.38	.067	1.70	.145	3.68	.270	6.86
.003	.08	.016	.40	.080	2.03	.150	3.81	.280	7.11
.004	.10	.020	.51	.100	2.54	.170	4.31	4.960	125.98
.005	.13	.025	.64	.120	3.05	.190	4.83	5.200	132.08
.007	.18	.040	1.02	.125	3.18	.225	5.72	5.440	138.18

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.005 (.13 mm) inch for three place decimals, ±.01 (.3 mm) inch for two place decimals and ±2° for angles.
4. Flash around blade contacts not to exceed .010 (.25 mm) inch from the surface of the insulator and shall be firmly attached.
5. Pin shield shall be marked with contact numbers 1, 50, 51 and 100 in contrasting color ink .06 (1.5 mm) inch high.
6. Single or multiple piece construction is optional provided that the end requirements of the drawing are met. It is permissible for the connector body to be installed in either an aluminum or thermoplastic shell which provides the pin shield and key pin hole features. The thermoplastic shall be type GLT-30F.
7. Dimension prior to solder coating.
8. Dimensions apply after plating.
9. See Table I for key pin hole size and body material.
10. Contact engagement length.
11. Shaded area shall be solder coated.
12. This connector is not to be used with ceramic circuit boards.
13. Connector shall be marked in approximate location shown with part number, manufacturer's number and date code with contrasting ink in characters .06 (1.5 mm) inch high.

FIGURE 1. Dimensions and configurations - continued

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REQUIREMENTS:

Dimensions and configuration: See Figure 1 and Table I.

Materials: Contacts shall be either copper alloy sheet or strip, UNS Number C72500, temper 1/4 hard minimum, in accordance with ANSI/ASTM B 122 or brass, QQ-B-613, alloy 260, temper 1/2 hard.

Connector body shall be either polyetherimide, Type GLT-20F or GLT-30F, molded thermoplastic in accordance with MIL-M-24519 or molding plastic, diallyl phthalate resin, GL filler, MIL-M-14, Type GDI-30F or Type SDG-F.

The pin shield shall be aluminum alloy 5052, QQ-A-250/8, H34, or glass-epoxy, GF-0160-00/00-0-1-X, MIL-P-13949 or polyetherimide, Type GLT-20F or GLT-30F, molded thermoplastic in accordance with MIL-M-24519. Pin shield thickness for molded thermoplastic shall be 0.020 (.51 mm) inch minimum.

Plating: Contacts shall be finished on all exposed surfaces. The contact engagement length indicated shall be finished with gold plate, MIL-G-45204, Type II, Grade C, Class 1, 0.000050 (.00013 mm) inch minimum thickness over nickel plating, QQ-N-290, Class 1, 0.000050 (.00013 mm) inch minimum thickness.

Selectively solder coat the leads as shown in the shaded area with QQ-S-571, SN60 or SN63, to a thickness of 0.0005 (.013 mm) inch to 0.003 (.08 mm) using flux, MIL-F-14256, Type RMA.

All aluminum alloy parts shall be anodic coated in accordance with MIL-A-8625, Type III, Class 2, black.

Key pin holes: For M28754/42-02 material, .095 ± .001 (2.41 ± .03 mm) inch diameter x .150 (3.81 mm) inch deep.

For M28754/42-01 material, .114 $^{+.000}_{-.005}$ (2.90 $^{+.00}_{-.13}$ mm) inch diameter x .150 (3.81 mm) inch deep.

C'bore .134 $^{+.005}_{-.000}$ (3.40 $^{+.13}_{-.00}$ mm) inch dia x .020 $^{+.005}_{-.000}$ (.51 $^{+.13}_{-.00}$ mm) inch deep

Three equally spaced 0.010 (.25 mm) inch rad beads x .150 (3.81 mm) inch deep around the periphery of the .114 (2.90 mm) inch diameter shall be utilized.

Electrical requirements:

Dielectric withstanding voltage: The connector shall meet the requirements of MIL-STD-1344, Method 3001, for 600 VAC (RMS) 60 Hz, or 850 VDC. The point of measurement shall be between adjacent contacts.

Low level (dry circuit): Voltage drop shall not exceed 36 millivolts. Wire size is not applicable.

Contact resistance: Voltage drop shall not exceed 36 millivolts. Current Rating; 3 Amp

Temperature rating: -55°C to +125°C

Voltage rating: 300 Volts, AC (RMS), at sea level

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Durability: In accordance with MIL-C-28754 and shall not exceed 36 millivolts.

Solderability: In accordance with MIL-C-28754.

Military part number: M28754/42- (Dash Number from Table I)

TABLE I

DASH NUMBER	KEY PIN HOLE SIZE INCH (MM)	BODY MATERIAL
01	$.114\ ^{+.000}_{-.005}$ $(2.90\ ^{+.00}_{-.13})$	Type GDI-30F or SDG-F
02	.095 ± .001 (2.41 ± .03)	Type GLT-20F or GLT-30F

Warning: The GLT-20F or GLT-30F may be susceptible to stress cracking when exposed to certain cleaning solvents.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Qualification information: The connector shall meet the qualification test of MIL-C-28754, Table IV, Type IV, except for Subgroup IV, Method Paragraph 4.7.18, 4.7.19 and 4.7.20.

Number of units to be inspected: Twelve (12) of each dash number to be qualified shall be inspected.

Supersession data: Supersedes Naval Electronic Systems Command Drawing Number 0102-691-2 and -7, Code Identification Number 28687.

Custodians:
Navy - EC

Review Activities:
Navy - MC

User Activities:
Navy - AS

Preparing Activity:
Navy - EC

(Project: 5935-N252-01)

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